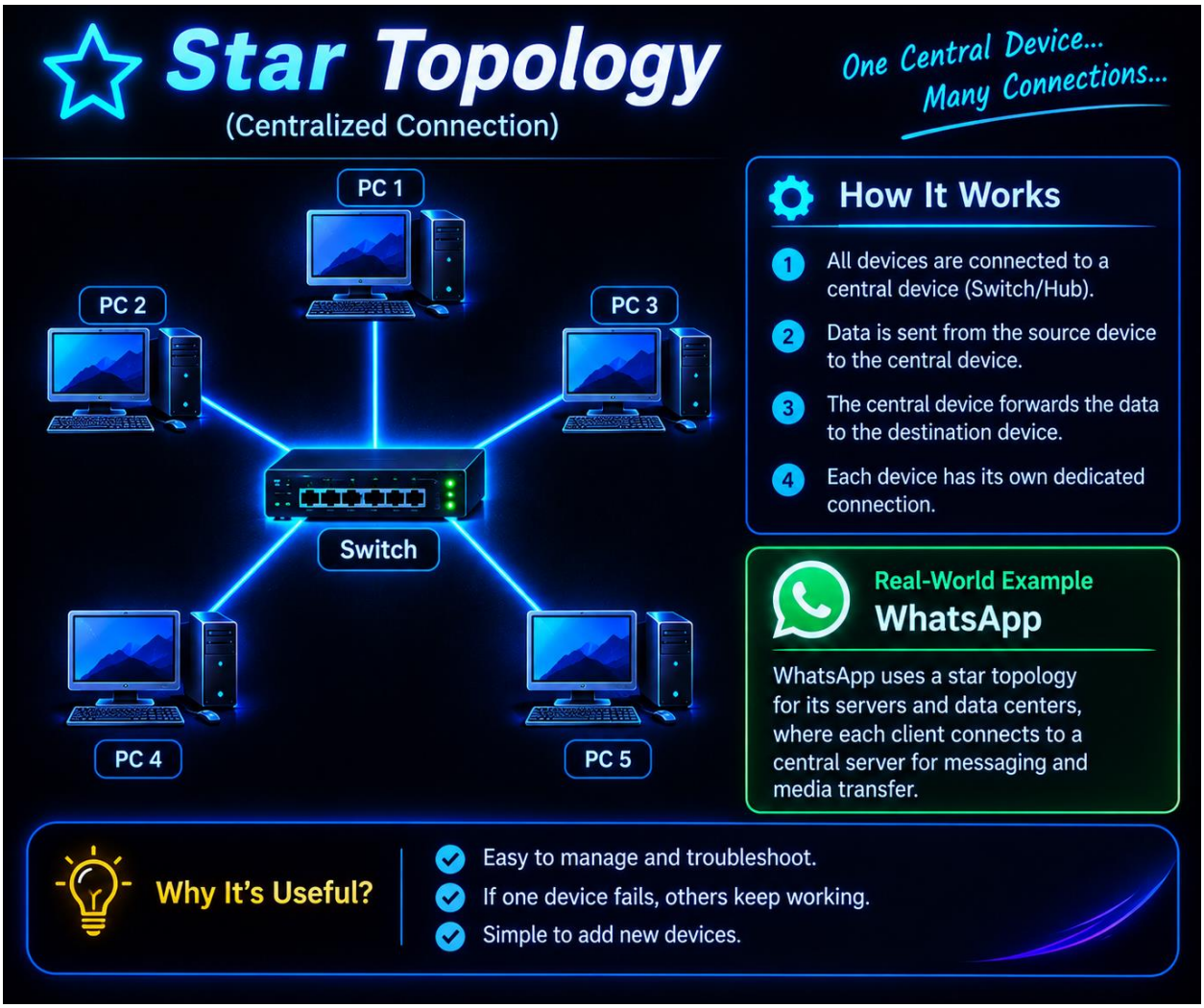
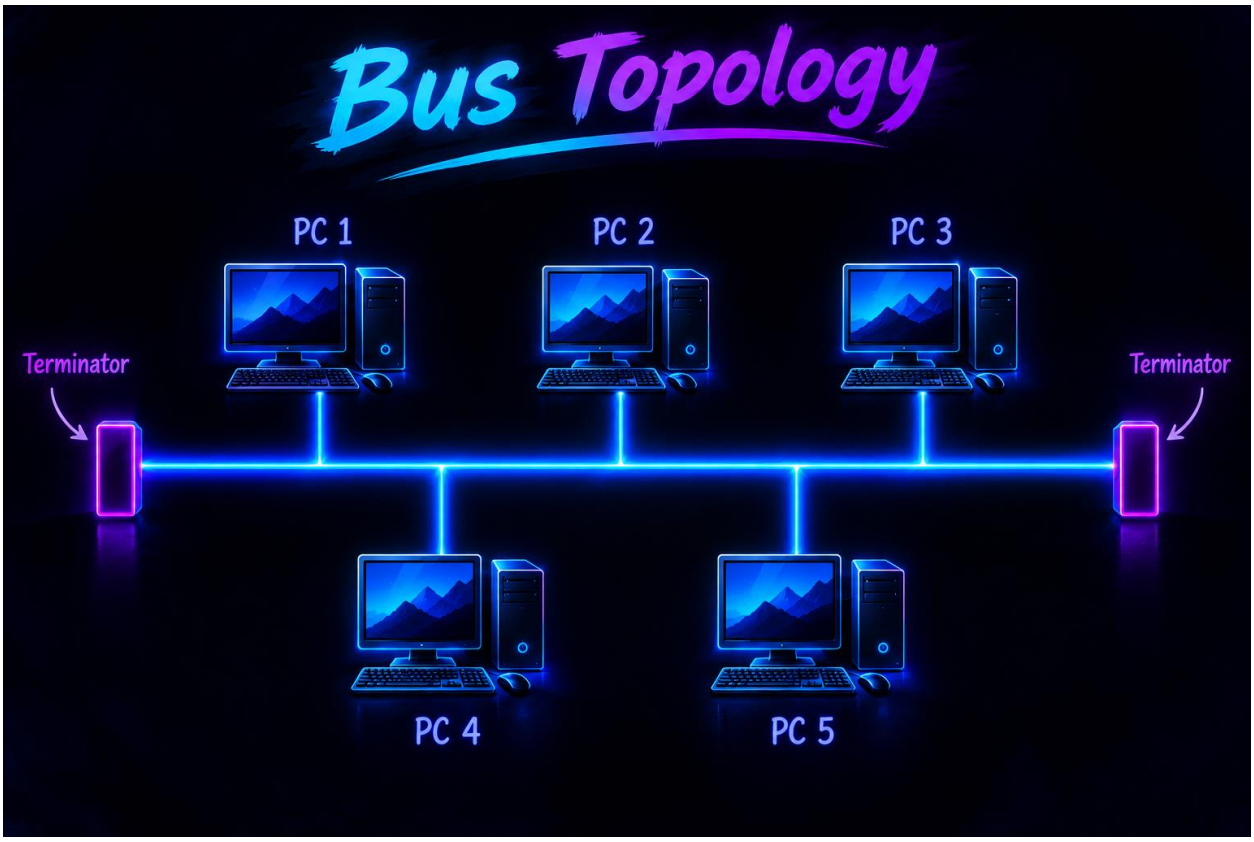
1 Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.



Example: WhatsApp

WhatsApp uses a centralized server system to connect users and transfer messages, photos, videos, and other files. Each user's device communicates with WhatsApp's servers, which help deliver data to the correct recipient. This makes communication fast and convenient.

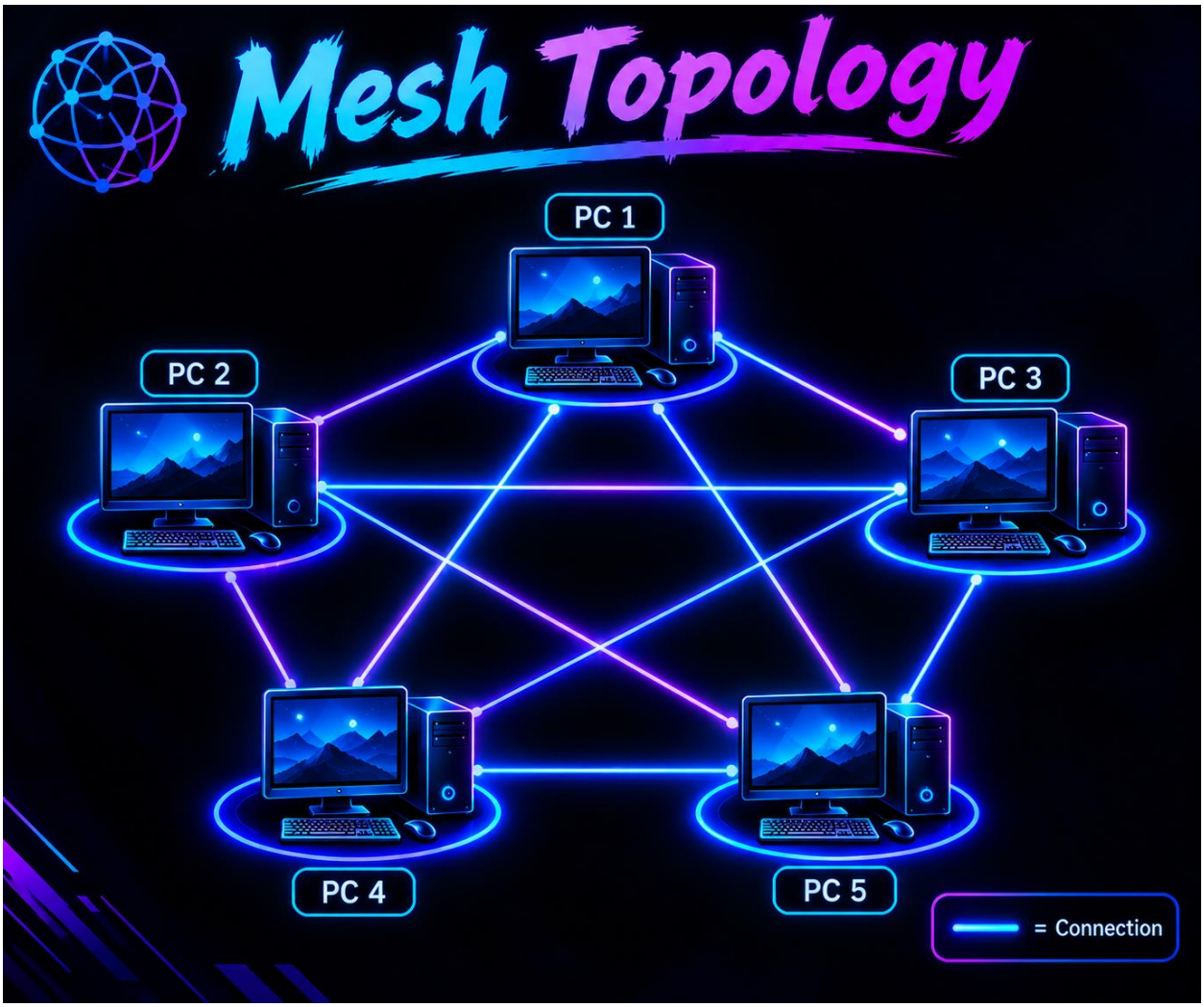
2 Bus topology :



Example: Small Office Network

A small office can use bus topology to connect multiple computers through a single main cable. It uses less cable and is cost-effective, but if the main cable fails, the entire network may stop working.

3 Mesh Topology :



Example: PUBG Mobile

PUBG Mobile requires fast and reliable communication between game servers and players. Mesh topology provides multiple communication paths, so if one connection fails, another path may help maintain communication. However, it can be expensive and complex to manage.

2. Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for

'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case & Example
Star	1. Easy to install. 2. Easy to troubleshoot. 3. Easy to expand. 4. One PC failure does not affect others.	1. Requires more cables. 2. Needs a central switch. 3. Switch failure stops the network. 4. Installation cost can be high.	Office Network — Flipkart
Bus	1. Low cost. 2. Uses less cable. 3. Easy to set up for small networks. 4. No central switch required.	1. Main cable failure stops the network. 2. Troubleshooting is difficult. 3. Performance decreases with heavy traffic. 4. Limited expansion.	Small LAN — Small office
Ring	1. Data travels in an organized path. 2. Predictable data transmission. 3. No central switch is required in a basic ring. 4. Can	1. A single link failure can disrupt a basic ring. 2. Adding devices can interrupt the network. 3. Troubleshooting	Ring LAN — Specialized office network

	reduce data collisions in token-based rings.	can be difficult. 4. Repair and expansion can be costly.	
Mesh	1. Highly reliable. 2. Multiple paths for data. 3. Can continue working after a link failure. 4. Provides good redundancy.	1. Expensive to install. 2. Requires many cables or links. 3. Complex to configure. 4. Difficult to maintain in large networks.	Reliable Network — PUBG-style multiplayer gaming infrastructure

3. Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

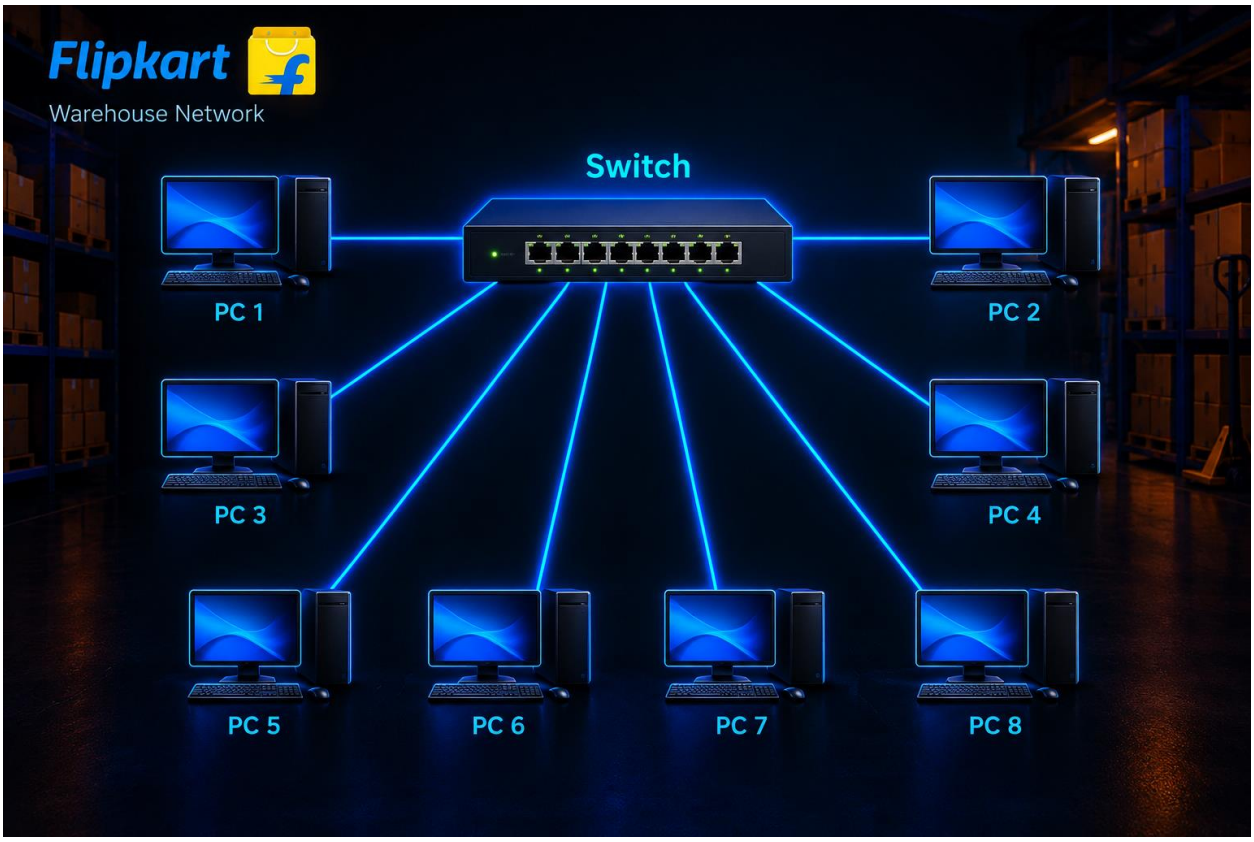
Topology: Mesh Topology

- Mesh topology provides multiple paths for data, helping players communicate quickly.

- It is reliable because another path may work if one connection fails.
- It helps maintain smooth gameplay and reduces connection problems.
- However, it is expensive to set up and maintain.

4 Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.



Reasoning:

- Cost: Star topology is affordable and suitable for 8 computers.
- Troubleshooting: If one computer or cable fails, the other computers continue working.
- Future Expansion: More computers can be easily connected to the central switch.